Mastering Docker: Essential Commands for Daily Development

Docker helps you run applications smoothly on any computer without messy setup issues. Learning these core commands will save you hours of frustration and make you feel like a pro.

What you'll learn:

1. Downloading and Running Your First App
2. Checking What Is Alive
3. Debugging with Container Logs
4. Stopping and Cleaning Up

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Downloading and Running Your First App

Before you can run anything, Docker needs the app files. Think of 'pull' like downloading a game from an app store. Once downloaded, 'run' starts the game inside a safe sandbox on your computer. The -d flag lets it run quietly in the background, and -p connects your computer's port to the app's port.

```
$docker pull nginx
 docker run -d -p 8080:80 nginx
```

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Checking What Is Alive

When you have multiple apps running, it is easy to lose track. The 'ps' command lists every active container currently running on your machine. If you need to see everything including stopped apps, just add the -a flag to the end.

```
$docker ps
 docker ps -a
```

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Debugging with Container Logs

When an app fails inside Docker, you cannot just open a normal terminal inside it easily. Instead, you look at the logs. This command streams the app's console output directly to your screen so you can spot error messages instantly.

```
$docker logs -f <container-id-or-name>
```

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Stopping and Cleaning Up

Containers keep running until you explicitly tell them to stop. Use the stop command with the container ID. Over time, old containers and unused data pile up and eat disk space, which is easily cleaned up with the system prune command.

```
$ docker stop <container-id>
docker system prune -f
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

Key Takeaways

- + Always give your containers meaningful names using --name so you can find them easily.
- + Use 'docker logs' first whenever a containerized app refuses to load.
- + Run 'docker ps' regularly to keep track of what background resources are active.
- + Keep your Dockerfiles simple and add comments to explain each step.
- + Practice typing these commands daily until they become second nature.

Watch Out For

- ! Forgetting to stop background containers, which wastes laptop memory and battery.
- ! Panicking when an app fails instead of checking the container logs first.
- ! Confusing container IDs with image names when running commands.

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